Honours Linear Algebra - Week 5

12 and 14 October 2026

12 Oct. / Rank; change of basis

$T : V \to W$, finite dimensions:

$$r \le \min\{\dim V, \dim W\}, \qquad r + \dim \ker T = \dim V.$$

Full rank means $r = \min\{\dim V, \dim W\}$. $T(x) = b$: solvable iff $b \in \operatorname{Im} T$; then solutions $= x_p + \ker T$.

Thm.2.32. For square $A \in M_n(K)$, equivalent:

$$\begin{aligned} A \text{ invertible} &\iff \operatorname{rank} A = n \iff \operatorname{rref}(A) = I_n \\ &\iff \ker A = \{0\} \iff \operatorname{Col}(A) = K^n \\ &\iff \text{columns of } A \text{ form a basis of } K^n. \end{aligned}$$

Thm.2.33–35. Ordered bases $\mathcal{B} = (b_i)$, $\mathcal{C} = (c_i)$:

$$P_{\mathcal{C}}^{\mathcal{B}} = ([b_1]_{\mathcal{C}} \ \cdots \ [b_n]_{\mathcal{C}}), \qquad [v]_{\mathcal{C}} = P_{\mathcal{C}}^{\mathcal{B}}[v]_{\mathcal{B}}.$$

Source upstairs, target downstairs. It is the identity map in two coordinate systems.

$$\left(P_{\mathcal{C}}^{\mathcal{B}}\right)^{-1} = P_{\mathcal{B}}^{\mathcal{C}}, \qquad P_{\mathcal{D}}^{\mathcal{C}} P_{\mathcal{C}}^{\mathcal{B}} = P_{\mathcal{D}}^{\mathcal{B}}.$$

For bases of K^n, put $S_{\mathcal{B}} = (b_1 \ \cdots \ b_n)$ in standard coordinates. Then

$$v = S_{\mathcal{B}}[v]_{\mathcal{B}} = S_{\mathcal{C}}[v]_{\mathcal{C}} \quad \Rightarrow \quad P_{\mathcal{C}}^{\mathcal{B}} = S_{\mathcal{C}}^{-1} S_{\mathcal{B}}.$$

Ex.2.23. $\mathcal{B} = ((1,1)^T, (1,-1)^T)$, $\mathcal{C} = ((1,0)^T, (1,1)^T)$:

$$P_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 1 & -1 \end{pmatrix}.$$

Column check: $b_1 = c_2$, $b_2 = 2c_1 - c_2$.

Similar matrices

Thm.2.36–37. $A = [T]_{\mathcal{B}}$, $B = [T]_{\mathcal{C}}$, $Q = P_{\mathcal{C}}^{\mathcal{B}}$:

$$BQ = QA \quad \Rightarrow \quad B = QAQ^{-1}.$$

Same linear map, different bases. Similarity is an equivalence relation; rank and trace are unchanged. The converse from these invariants alone is false.

A nicer basis; determinant axioms

Ex.2.24. In standard coordinates,

$$A = \frac{1}{5}\begin{pmatrix} 14 & -2 \\ -2 & 11 \end{pmatrix}, \qquad S = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}, \qquad AS = S\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}.$$

Thus $[T]_{\mathcal{B}} = S^{-1}AS = \operatorname{diag}(2,3)$ for $\mathcal{B} = ((1,2)^T, (-2,1)^T)$. Here S sends **new coordinates to standard coordinates**, so the inverse is on the left.

Multilinear: linear in each slot separately, holding the others fixed.

$$F(\ldots, au + bv, \ldots) = aF(\ldots, u, \ldots) + bF(\ldots, v, \ldots).$$

Symmetric: swaps leave F unchanged. Alternating: repeated inputs give zero; a swap changes sign. The swap-sign rule alone implies alternating when $\operatorname{char} K \neq 2$.

Example: $F(u,v) = u^T M v$ is bilinear; symmetric for $M^T = M$, alternating for skew-symmetric M over $\mathbb{R}$ or $\mathbb{C}$.

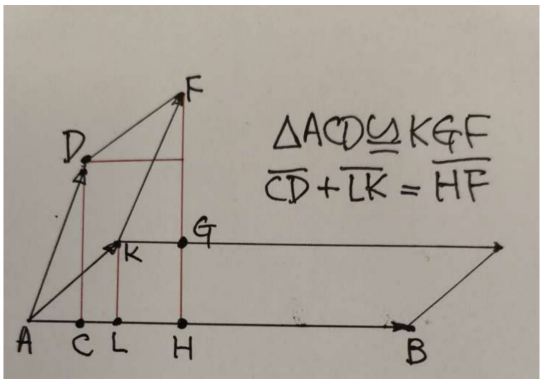
Def.3.1. $\det(a_1, \ldots, a_n)$ is multilinear and alternating in its columns, normalized by $\det I_n = 1$.

$$\begin{aligned} \det(\ldots, ca_i, \ldots) &= c \det A, \\ \det(\ldots, a_i + ca_j, \ldots, a_j, \ldots) &= \det A \quad (i \neq j), \\ \det(\ldots, a_j, \ldots, a_i, \ldots) &= -\det A. \end{aligned}$$

Dependent columns $\Rightarrow \det A = 0$: express one column through the others; every expanded term has a repetition.

Prop.3.2. Triangular A: $\det A = \prod_i a_{ii}$. Also $\det(cA) = c^n \det A$; scaling the whole matrix scales every column.

Geometry over $\mathbb{R}$. Determinant = oriented volume; ordinary volume = $|\det A|$. For a fixed base, signed heights add:



14 Oct., p.26. $CD + LK = HF$: area additivity in one slot.

14 Oct. / Existence and uniqueness

Permutations. S_n: bijections of $\{1,\dots,n\}$. A transposition swaps two entries; every permutation is a product of transpositions. Its parity is well-defined:

$$\operatorname{sgn}(\sigma) = (-1)^{\#\text{ transpositions}}, \qquad \operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau).$$

Thm.3.3 / Leibniz.

$$\boxed{\det A = \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) \prod_{j=1}^{n} a_{\sigma(j),j}.}$$

One entry from each row and each column, with its permutation sign.

Uniqueness: for any alternating multilinear D, expand $a_j = \sum_i a_{ij}e_i$:

$$\begin{aligned} D(a_1,\dots,a_n) &= \sum_{i_1,\dots,i_n} \left(\prod_j a_{i_j j}\right) D(e_{i_1},\dots,e_{i_n}) \\ &= \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) \left(\prod_j a_{\sigma(j),j}\right) D(e_1,\dots,e_n). \end{aligned}$$

Repeated indices vanish; only permutations survive. Hence $D(A) = D(I)\det A$; normalization fixes $D(I) = 1$.

Existence: the displayed formula is multilinear. Equal columns pair equal products with opposite signs, giving zero; at I only the identity permutation survives.

Ex.3.2–4.

$$\det(a) = a, \qquad \det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc,$$

$$\det\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = aei + bfg + cdh - afh - bdi - ceg.$$

The cyclic diagonal shortcut is only for 3×3.

Ex.3.1 / column operations.

$$\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 8 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 4 & -3 & -6 \\ 7 & -6 & -13 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 4 & -3 & 0 \\ 7 & -6 & -1 \end{vmatrix} = 3.$$

$C_2 \leftarrow C_2 - 2C_1$, $C_3 \leftarrow C_3 - 3C_1$, then $C_3 \leftarrow C_3 - 2C_2$.

Products, transpose, blocks

Cor.3.4–7.

$$\det(AB) = \det A \det B,$$
$$A \text{ invertible } \iff \det A \neq 0, \qquad \det(A^{-1}) = (\det A)^{-1},$$
$$B = Q^{-1}AQ \Rightarrow \det B = \det A.$$

Product proof: fix A. $D(B) = \det(Ab_1, \ldots, Ab_n)$ is alternating multilinear; uniqueness gives $D(B) = D(I)\det B = \det A \det B$.

Invertible $\Rightarrow \det A \det A^{-1} = 1$; singular $\Rightarrow$ dependent columns $\Rightarrow \det A = 0$. Therefore $\det T := \det[T]_{\mathcal{B}}$ is independent of the basis.

Thm.3.11. $\det A^T = \det A$. In Leibniz, replace σ by σ^{-1}; sign unchanged.

Thus row and column operations obey the same rules:

operation	new determinant
$R_i \leftrightarrow R_j$	$-\det A$
$R_i \leftarrow cR_i$	$c \det A$
$R_i \leftarrow R_i + cR_j$ $(i \neq j)$	$\det A$

Elementary matrices: row operations EA, column operations AE; determinants $-1, c, 1$, respectively. Triangularize and keep the swap/scaling factors.

Cor.3.8 / block triangular. Square diagonal blocks:

$$\det \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = \det A \det D.$$

Proof: factor as $\begin{pmatrix} I & B \\ 0 & D \end{pmatrix} \begin{pmatrix} A & 0 \\ 0 & I \end{pmatrix}$; clear B by column additions. No invertibility assumption needed.

Schur complement. For $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$,

$$A \text{ invertible}: \quad \det M = \det A \det(D - CA^{-1}B),$$
$$D \text{ invertible}: \quad \det M = \det D \det(A - BD^{-1}C).$$

First case: multiply on the left by $\begin{pmatrix} I & 0 \\ -CA^{-1} & I \end{pmatrix}$ (determinant 1), making the bottom-left block zero. Keep the order of the factors.

Cofactors; solving equations

Thm.3.14 / Laplace. Delete row i, column j to obtain A_{ij}; cofactor $C_{ij} = (-1)^{i+j} \det A_{ij}$.

$$\det A = \sum_{j=1}^{n} a_{ij} C_{ij} \quad \text{(fixed row } i\text{)}, \qquad \det A = \sum_{i=1}^{n} a_{ij} C_{ij} \quad \text{(fixed column } j\text{)}.$$

Proof: expand $a_j = \sum_i a_{ij} e_i$. Replacing column j by e_i leaves $(-1)^{i+j} \det A_{ij}$ after clearing row i and moving the 1 to the corner.

Ex.3.5. Choose the first column (two zeros):

$$\begin{vmatrix} 0 & 2 & 3 \\ 1 & 0 & 1 \\ 0 & 3 & 2 \end{vmatrix} = - \begin{vmatrix} 2 & 3 \\ 3 & 2 \end{vmatrix} = -(4 - 9) = 5.$$

Thm.3.15 / Cramer. $\det A \neq 0$, $Ax = b$:

$$x_i = \frac{\det A_i}{\det A}, \qquad A_i = (a_1, \ldots, a_{i-1}, b, a_{i+1}, \ldots, a_n).$$

Proof: $b = \sum_j x_j a_j$; expand column i. All terms repeat a column except $x_i \det A$. Geometrically: a ratio of oriented volumes.

Ex.3.6.

$$A = \begin{pmatrix} 2 & 0 & 2 \\ 0 & 3 & 2 \\ 0 & 1 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \det A = 2, \quad (\det A_1, \det A_2, \det A_3) = (-13, -8, 14).$$

$$x = \frac{1}{2} \begin{pmatrix} -13 \\ -8 \\ 14 \end{pmatrix} = \begin{pmatrix} -13/2 \\ -4 \\ 7 \end{pmatrix}.$$

Useful closed formula; elimination is preferable for large systems.

Thm.3.16 / adjugate. Form the cofactor matrix $C = (C_{ij})$ and transpose it:

$$\operatorname{adj}(A) = C^T, \qquad A^{-1} = \frac{\operatorname{adj}(A)}{\det A} \quad (\det A \neq 0).$$

Column j of A^{-1} solves $Ax = e_j$; Cramer + Laplace gives $(A^{-1})_{ij} = C_{ji} / \det A$.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \quad (ad - bc \neq 0).$$